Price–Volume Information in Network Herding Models: Identification, Market Tests and Risk Diagnostics

Econophysics Risk Lab contributors

Collective dynamics, inference and market risk

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Abstract

Collective trading can amplify market fluctuations, yet aggregation can erase the mechanism information needed for inference. We study this problem in a finite Erdős–Rényi herding model with an explicit joint price–volume observation rule. Exact moments connect group-size power sums to observable risk and yield parameter estimators when population and unit order size are known. Ten independent recovery studies place finite-size moment inversion alongside supervised estimators, while exact observational equivalence shows why complete aggregate observations can leave internal network edges unidentified. A dimensionless joint-moment restriction provides a market test: the BTCUSDT 2025 H2 estimate is -8.181, outside the fixed model's positive range. Controlled activity and response states then separate same-period information from temporal memory and predictive value. Historical next-day volatility forecasts and sector-portfolio diagnostics evaluate practical uses of observable information, with explicit statistical references and source-linked outputs. The resulting analysis connects generative mechanisms, identification, empirical compatibility and risk diagnosis, while retaining the conditions under which each inference holds.

1 Introduction

Aggregate price and volume compress the decisions of many market participants into a small set of observations. Coordination can amplify fluctuations, but similar aggregate distributions can arise from different microscopic structures. A useful mechanism model must therefore answer both a forward question, how interactions affect observations, and an inverse question, which features of those interactions observations can recover.

The Cont–Bouchaud model relates shared actions within random groups to aggregate market fluctuations[1]. Adaptive models such as Lux–Marchesi add switching and feedback to the dynamics[2]. Quantitative calibration studies emphasize the distinction between reproducing market patterns and identifying the parameters that generate them[6, 7]. We use a finite Erdős–Rényi (ER) coordination model with an explicit price–volume observation rule to study that distinction. Model context, including bubble and crash mechanisms, is developed in Appendix A.

The analysis follows three questions. First, which exact moment relations connect coordination to observable risk, and which model parameters do they identify? Second, what information remains hidden even with the complete observation law, and do real market observations satisfy the model's necessary restrictions? Third, how does persistent activity or price response change the information available for forecasting and risk diagnosis? Finite-graph calculations and controlled simulations establish the mechanism results; Bitcoin observations test the fixed interface; historical forecasting and sector-portfolio diagnostics translate observable information into analysis tools.

2 Finite network model and observable moments

2.1 Agents, actions, and the observation interface

Each period draws $H \sim G(N, p)$, where $p = \rho/(N-1)$ and $N \geq 2$. Nodes are model traders, edges represent coordination, and a connected cluster C contains s_C traders. Each cluster shares

$\xi_C \in \{-1, 0, 1\}$ with probabilities $q/2, 1-q, q/2$. Actions are independent across clusters and independent of the graph. Throughout, q is the total activity probability. The implementation uses the equivalent per-direction probability $a = q/2$.

Each active trader submits a unit order $u > 0$. A simplified external counterparty absorbs all submitted orders, and the log-price response is linear. The observations are

$$r = \frac{\lambda}{N} \sum_C s_C \xi_C, \qquad V = u \sum_C s_C |\xi_C|, \tag{1}$$
$$S_j = \sum_C s_C^j, \qquad \lambda > 0, \quad 0 < q < 1.$$

Signed price effects can offset, whereas trading activity adds. Here V counts executed activity and λ is a model response scale. Table 1 defines the notation, and Fig. 1 illustrates the observation rule.

Table 1. Model notation. A connected group shares one action within a period.

Symbol	Definition
N, ρ	Number of traders and expected ER degree, with $p = \rho/(N-1)$
C, s_C, S_j	Connected group, its size, and the power sum $S_j = \sum_C s_C^j$
q, ξ_C	Total activity probability and shared sell/inactive/buy action
u, λ	Unit order size and positive log-price response scale
r, V	Log return and total executed model volume per period
$A_N(\rho)$	Expected squared-size sum $\mathbb{E}S_2$ under the finite ER graph law

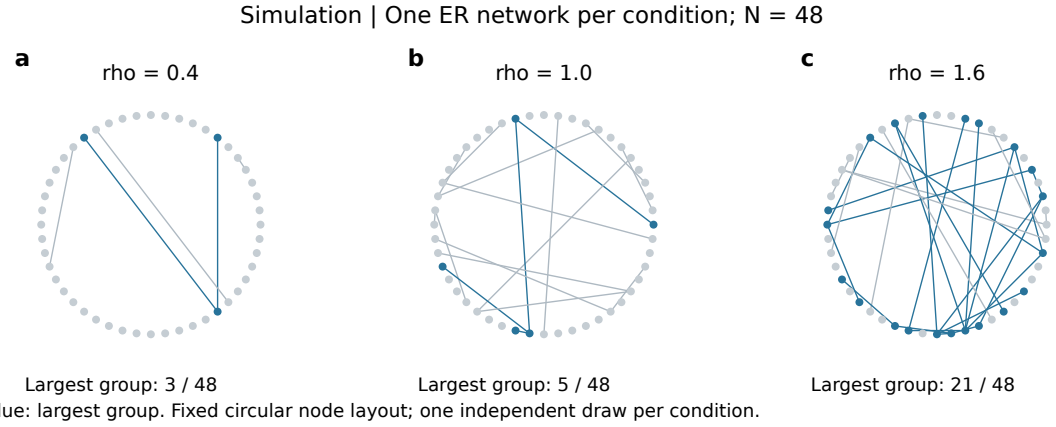


Figure 1. Shared group actions generate net price effects and total trading activity through the same execution rule. Nodes are model traders and connected groups coordinate their actions. [F01]

2.2 Exact moments and their mechanistic use

Zero-mean cluster actions eliminate cross terms in return variance. Conditional mean volume does not depend on the graph. Writing $A = \mathbb{E}S_2$ gives

$$\mathbb{E}V = quN, \tag{2}$$
$$\mathrm{Var}(r) = q(\lambda/N)^2 A,$$
$$\mathrm{Var}(V) = q(1-q)u^2 A.$$

Equal mean volume therefore need not imply equal risk. At fixed N, q, u, λ, a larger sum of squared cluster sizes increases return variance. For 100 independent traders, $S_2 = 100$; ten clusters of ten traders give $S_2 = 1000$. Variance differs tenfold in this model example while mean volume remains unchanged. The tenfold ratio follows from the specified group sizes and fixed model parameters.

Higher-order relations retain the connection between collective activity and large price changes:

$$\mathrm{Cov}(r^2, V) = u(\lambda/N)^2 q(1-q)\mathbb{E}S_3, \tag{3}$$
$$\kappa_3(V) = u^3 q(1-q)(1-2q)\mathbb{E}S_3.$$

Signed r and V have zero population covariance, while $|r|$ and V are positively associated under this interface. Price direction and magnitude thus require separate analysis. The empirical analysis below tests stronger joint restrictions than this association alone.

Conditional on the graph, the joint characteristic function is

$$\mathbb{E}[e^{itr+izV} \mid H] = \prod_C [1 - q + qe^{izus_C} \cos(t\lambda s_C/N)]. \tag{4}$$

It depends only on the multiset of cluster sizes, directly supporting the identification boundary below. For fixed N, $|r| \leq \lambda$, so the mathematical tail is bounded. This bounded support sets the domain for the finite-range tail tests below.

Source tables and methods: `docs/theory/FOUNDATIONS.md`.

3 Percolation and finite return distributions

In the infinite sparse ER limit, degrees are Poisson(ρ) and $G_0(z) = G_1(z) = e^{\rho(z-1)}$. The edge-exploration generating function satisfies $H_1(z) = zG_1(H_1(z))$. A giant component emerges when the mean branching number crosses one. Its fraction satisfies

$$P_\infty = 1 - e^{-\rho P_\infty}, \qquad \rho_c = 1. \tag{5}$$

Below threshold, the expected cluster size of a uniformly selected node is $1/(1-\rho)$. In finite systems, S_2/N differs from the statistic excluding one largest component,

$$\chi_f = \frac{\sum_{C \neq C_{\max}} s_C^2}{\sum_{C \neq C_{\max}} s_C} \tag{6}$$

which is recorded as missing when no finite component remains.

We scanned $N = 128, 256, 512$, nine ρ values, and 40 graphs per cell, totaling 1,080 graphs. Figure 2 shows a smooth finite-size connectivity transition. Grid resolution and fluctuations affect peak positions. The infinite-system threshold organizes the finite-size comparison; the measured peak also depends on system size and grid resolution.

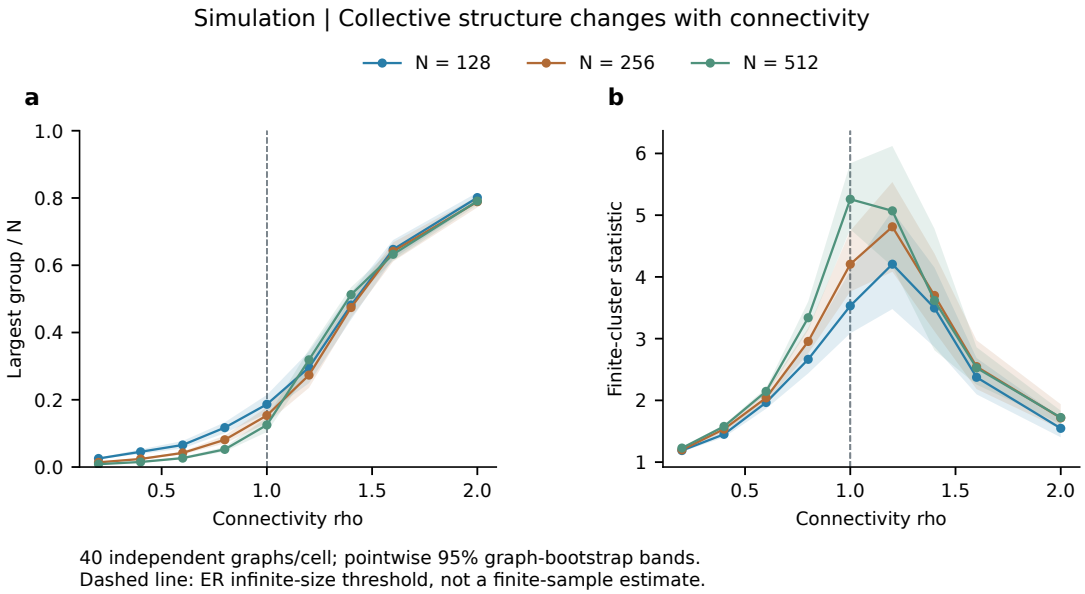


Figure 2. Finite-size connectivity transition for three population sizes. Each N/ρ cell contains 40 independent graphs. The vertical line marks the infinite sparse-ER threshold; finite-system peaks depend on size and grid resolution. Source: simulation_study_v1/phase_summary.csv. [F03]

The mean-field critical law $n_s \propto s^{-5/2}$ concerns cluster number density. Selecting a node uniformly instead weights cluster sizes by s. Returns sum signed cluster actions; activity, cancellation, and

bounded support affect their tails. For density $f(x) \propto x^{-\alpha}$, the complementary cumulative distribution (CCDF) exponent is $\alpha - 1$. These different sampling and aggregation operations require separate exponent definitions.

Five conditions each generated 8,192 periods, totaling 40,960 simulated intervals. Over the prespecified absolute net-order range $m = 4, \ldots, 128$, all three ρ conditions rejected the simple finite-support pure power law. The raw $p = 0.005$ is the resolution floor of 199 parametric-bootstrap draws; Holm adjustment across three tests gives $p = 0.015$. A better relative AIC under some conditions does not imply adequate absolute fit. The implementation follows maximum-likelihood, Kolmogorov–Smirnov (KS), and alternative-distribution principles[4], adapted to finite integer support.

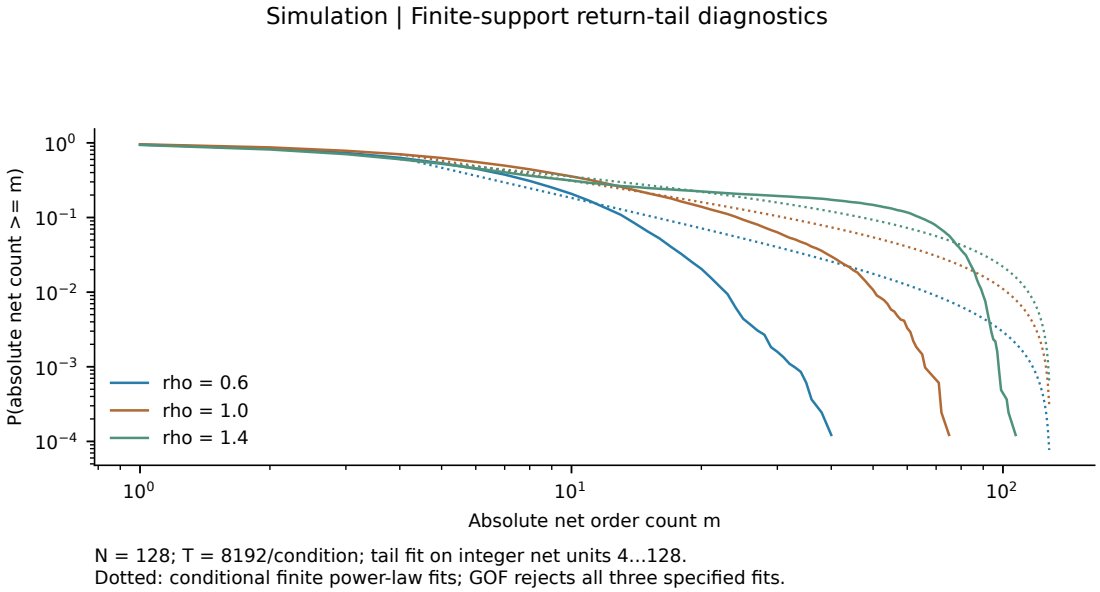


Figure 3. Return CCDFs and finite-support tail fits over the stated net-order range. The pure-power-law goodness-of-fit test rejects all three fitted connectivity conditions; relative AIC comparisons measure a separate property. [F05]

Collective amplification therefore survives even where this finite return distribution fails a pure-power-law fit. Cluster geometry and return tails answer different questions. Appendix B compares cluster distributions, population-size changes, and temporal aggregation.

4 Parameter recovery and observational equivalence

4.1 Finite-graph moment inversion

For known N, u under the fixed observation assumptions, population moments give

$$q = \frac{\mathbb{E}V}{uN}, \qquad A_N(\rho) = \frac{\mathrm{Var}(V)}{q(1-q)u^2}, \tag{7}$$
$$\lambda^2 = N^2u^2(1-q)\frac{\mathrm{Var}(r)}{\mathrm{Var}(V)}.$$

The ER mapping $A_N(\rho) = \mathbb{E}S_2$ is strictly increasing and therefore supports inversion. Finite-graph calculations use the classical connectivity recursion[5]:

$$C_s = 1 - \sum_{k=1}^{s-1} \binom{s-1}{k-1} C_k (1-p)^{k(s-k)},$$
$$\pi_{N,s} = \binom{N-1}{s-1} C_s (1-p)^{s(N-s)}, \tag{8}$$
$$A_N(\rho) = N \sum_s s\pi_{N,s}, \qquad C_1 = 1.$$

This classical recursion supplies the finite-size map needed by the moment estimator. The identity $A_N = N + N(N-1)\Pr(1 \leftrightarrow 2)$ connects it to pairwise connectivity. Coupling graphs by adding edges makes this probability strictly increasing in p for $N \geq 2$, which gives the population inverse its uniqueness. The implementation used 380 decimal digits, checked against higher precision and small-graph enumeration.

Ten independently regenerated studies produced 4,920 windows of 256 periods, totaling 1,259,520 simulated intervals. Each study used 252 training, 84 validation, and 84 test windows, plus 72 unseen intermediate-parameter windows. Moment inversion and classical radial-basis-function (RBF) methods share evaluation units. The return-only view contains return standard deviation, mean absolute return, the 95th and 99th absolute-return quantiles, and the zero-return fraction. The volume-only view contains volume mean, standard deviation and 95th percentile. Joint inputs combine these views and the Pearson correlation of volume with absolute return. Seeds and label metadata are excluded.

The supervised reference is kernel ridge regression with a Gaussian radial-basis kernel. For standardized observed features x, training labels $y_i = \rho_i$, and training mean $\bar{y}$,

$$\begin{aligned} K_{ij} &= \exp[-\gamma \|x_i - x_j\|^2], \\ \widehat{\rho}(x) &= \bar{y} + k(x)^\top (K + \alpha_K I)^{-1}(y - \bar{y}\mathbf{1}). \end{aligned} \tag{9}$$

Here $k_i(x) = \exp[-\gamma\|x - x_i\|^2]$ and α_K is the ridge penalty. Median imputation, missingness indicators and standardization are fitted on training windows only. Validation MAE selects the penalty and kernel width from the fixed configuration grid. For positive return standard deviation, the scale-invariant view divides the three return-magnitude summaries by that scale and retains the zero fraction and volume summaries; degenerate windows follow the imputation rule. Table 2 and Fig. 4 report the matched-scale results.

Table 2. Connectivity recovery in the specified ER family: mean MAE over ten independent studies, in units of ρ.

Method	Original grid	Unseen intermediate ρ
Return-only RBF	0.048660	0.045077
Joint RBF	0.027013	0.026882
Volume-only RBF	0.022196	0.036849
Scale-invariant joint RBF	0.032106	0.031013
Moment inversion	0.028372	0.027114

Each study regenerates training, validation, and evaluation windows. Source: depth_study_v1/summary_metrics.csv.

On the original grid, joint RBF reduced average MAE by 44.49% relative to return-only RBF. Volume-only RBF performed better than joint RBF in all ten studies. For unseen parameters, moment inversion achieved MAE 0.027114, compared with 0.026882 for joint RBF. Mechanistic formulas can therefore provide interpretable estimators within the specified family. The comparison establishes similar error levels in this experiment; no equivalence test was performed.

4.2 Price rescaling separates representation error from information loss

Each window undergoes $r \mapsto br$, with $b = 0.5, 1, 2$, while V remains unchanged. These are paired transformations of the same windows. Doubling the price response increases raw joint-RBF MAE from 0.027013 to 0.258133. The scale-invariant representation and moment method remain at 0.032106 and 0.028372, respectively. Figure 5 shows this representation trade-off. It establishes neither robustness to every market change nor non-identifiability from estimator failure.

4.3 Even complete observations need not reveal internal edges

The chain $\{12, 23, 34\}$ and star $\{12, 13, 14\}$ differ as four-node graphs, yet both form a single cluster of size four. Equation (4) gives identical complete joint price–volume laws. More passive

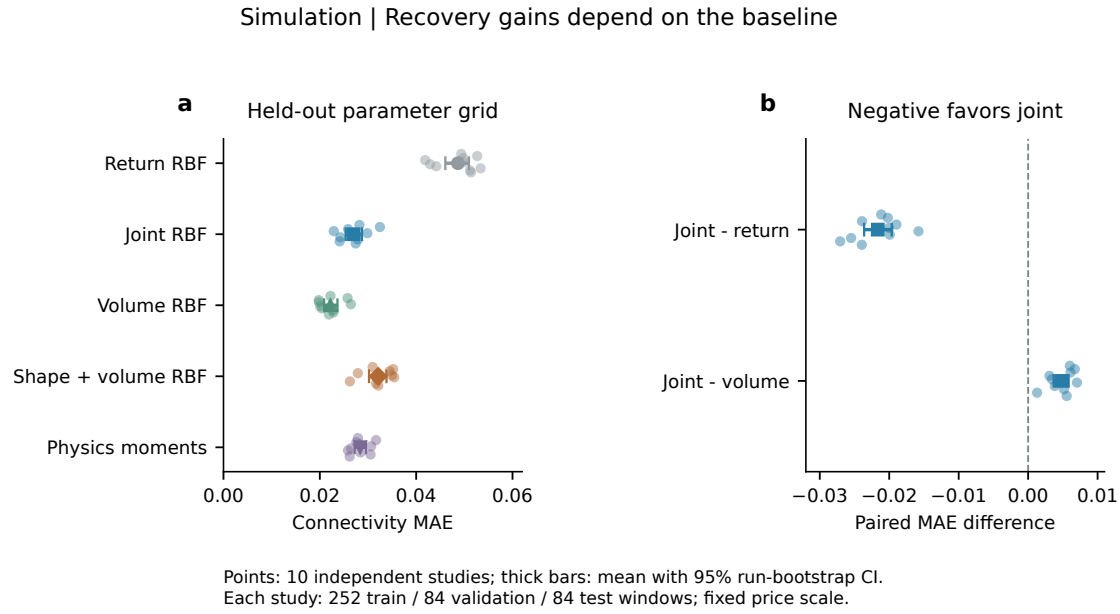


Figure 4. Parameter recovery across ten independently regenerated and retrained studies. Points are studies; summaries and intervals describe across-study variation. Table 2 gives grid and unseen-parameter MAE under the matched-scale design. [D01]

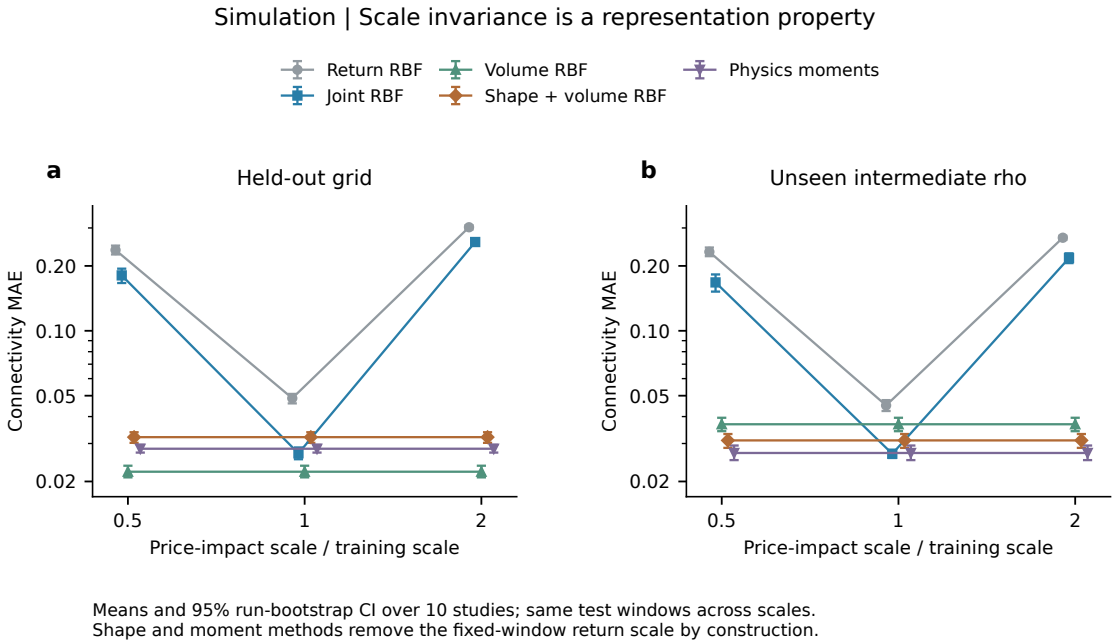


Figure 5. Price-response rescaling applied to the same evaluation windows in ten studies. Raw-return inputs change with scale, while moment inversion and scale-invariant inputs remove the specified window-level transformation by construction. Intervals resample studies. [D02]

observations of this kind cannot recover the internal edges. This is stronger than failure of a particular classifier. Exact certificates for general cluster mixtures are also included with proofs and rational checks.

Parameter recovery, structural equivalence, and information lost through feature compression require separate treatment. Financial ABM calibration studies likewise distinguish reproducing stylized facts from identifying mechanism parameters[6, 7]. The characteristic function, known-truth recovery experiments, and market restrictions thus address structural information, estimation accuracy, and model compatibility, respectively.

Source tables and methods: `docs/theory/DEPTH_ADDENDUM.md`.

5 Testing joint observations in Bitcoin

5.1 Data and price–volume comparison

The primary market case uses Binance Vision BTCUSDT spot five-minute bars[10]. The 2025 sample contains 105,120 bars and 365 complete UTC days, without filling or deletion. H1 provides calibration; H2 contains the original 184-day compatibility-test period. Returns are adjacent-close log changes, and bar volume is measured in BTC. Five-minute bars still compress the within-bar trade sequence. Figure 6 compares binned joint statistics and response proxies without treating volume as net capital flow.

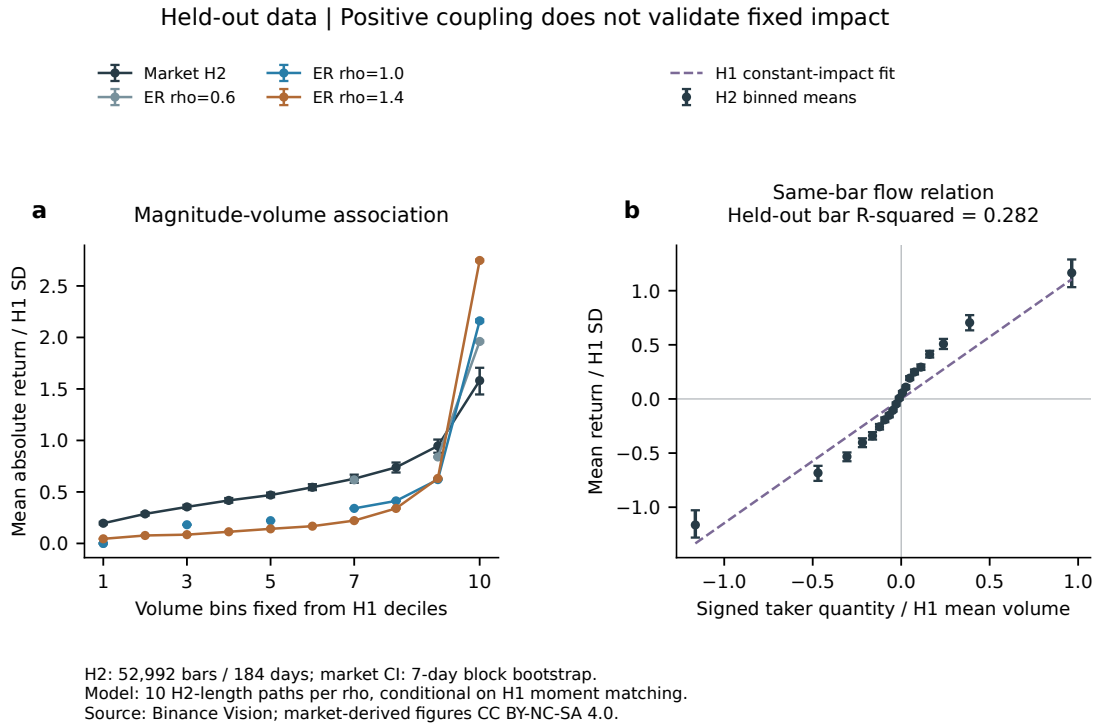


Figure 6. Bitcoin and model price–volume comparisons using H1-fixed bins. H2 magnitude–volume Spearman correlation is approximately 0.498, or 0.481 after seasonal adjustment. Empty model bins remain missing. Binance Vision, CC BY-NC-SA 4.0. [R01]

Appendix Fig. 16 compares market and model tails on a calibration-defined scale. Market data contain no true trader counts, clusters, or ρ labels. Simulation-trained outputs therefore cannot directly establish real-market criticality.

5.2 A joint necessary restriction derived from the mechanism

Eliminating scales from Eq. (3) gives a restriction for any graph distribution, beyond ER:

$$\Delta = \frac{1}{\mathbb{E}V}\left[\frac{\mathrm{Cov}(r^2, V)}{\mathrm{Var}(r)} - \frac{\kappa_3(V)}{\mathrm{Var}(V)}\right] = \frac{\mathbb{E}S_3}{N\mathbb{E}S_2} \in [1/N, 1]. \tag{10}$$

It requires fixed N, q, u, λ, symmetric independent actions, simplified execution, and linear response. Empirical calculations remove constant return drift and retain higher-moment and nonstationarity sensitivities.

For market H2, $\Delta = -8.181$, with an approximate 95% circular seven-day-block interval $[-11.998, -4.712]$. Figure 7 contrasts this estimate with the required positive range. The observed sign conflicts with the necessary range under the joint fixed-interface assumptions. All 12 calendar-month point estimates in 2025 are also negative, as are the checks excluding the

highest-volume or highest-realized-variance day. These source-table sensitivities support using the restriction as a model diagnostic. The test identifies a failure of the fixed interface, while the next section examines how changing activity and response affects that restriction.

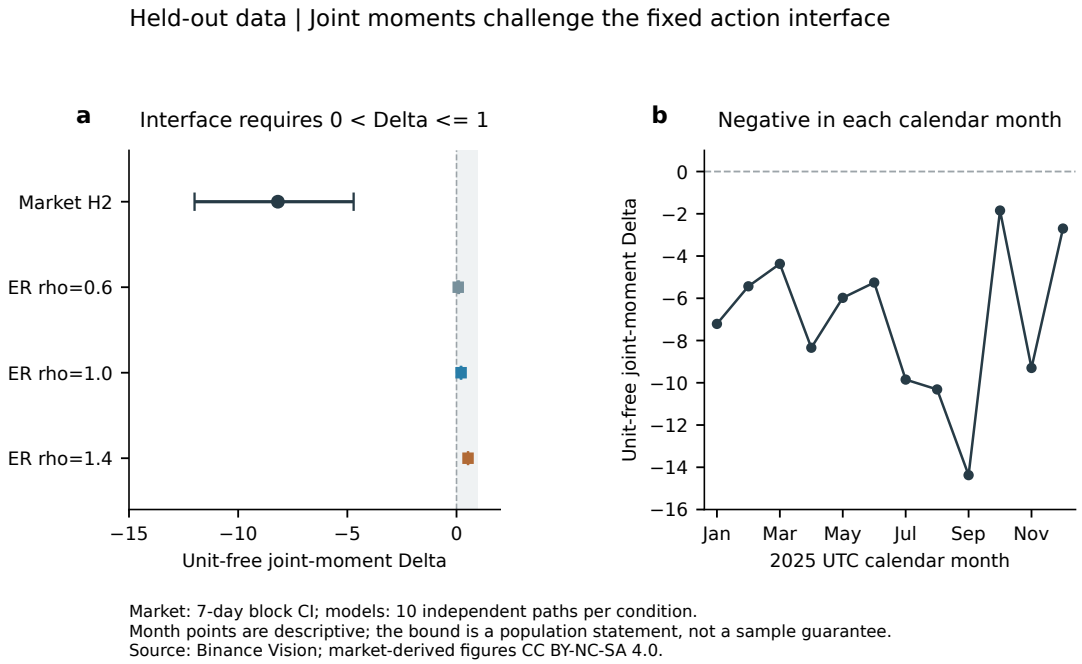


Figure 7. Joint-moment restriction for the fixed group-action interface and its BTC H2 estimate. The model requires a positive statistic. Market intervals use circular date-block resampling; monthly and extreme-day sensitivities are retained in the source tables. Binance Vision, CC BY-NC-SA 4.0. [R04]

5.3 Temporal structure supplies additional forecasting information

With fixed parameters and independently redrawn graphs and actions,

$$\mathbb{E}[r_{t+1}^2 \mid \mathcal{F}_t] = q(\lambda/N)^2 \mathbb{E}S_2 \tag{11}$$

is constant. This static baseline does not itself provide a date-varying next-period risk forecast. Market H2 has lag-one magnitude autocorrelation 0.344, signed-return autocorrelation about −0.021, and a one-day aggregation variance ratio about 0.749. Temporal information therefore supplies an additional axis of mechanism comparison beyond the same-period cluster geometry.

6 Temporal states and forecasting information

To isolate temporal information, the controlled state extension keeps $N = 128, \rho = 1$ and independent graphs, while activity q_t or response λ_t follows an exogenous temporal state. Four arms each use ten paths of 8,192 periods, totaling 327,680 observations. Population mean volume and return second moments match across arms. Persistent activity produces magnitude and volume memory; persistent independent response can produce magnitude memory with nearly absent volume memory. Shuffling complete (r, V) pairs preserves the same-period empirical joint distribution while removing temporal memory. Figure 9 separates these informational roles.

Let $m_k = \mathbb{E}(q^k)$ and $A = \mathbb{E}S_2$, with activity independent of the graph. The total-variance identity gives

$$\mathrm{Var}(V) = u^2[(m_1 - m_2)A + N^2(m_2 - m_1^2)]. \tag{12}$$

The second term arises from changing activity, preventing interpretation of all volume variance as cluster structure. A complementary analytical construction, developed after the simulation comparison, sets $\lambda(q)^2 = \lambda_0^2 \bar{q}/q$, keeping $\mathbb{E}[r_{t+1}^2 \mid \mathcal{F}_t] = (\lambda_0/N)^2 \bar{q} A$ constant. Volume may reveal activity states without improving this forecasting target. Its population $\Delta \approx -2.16348$ shows

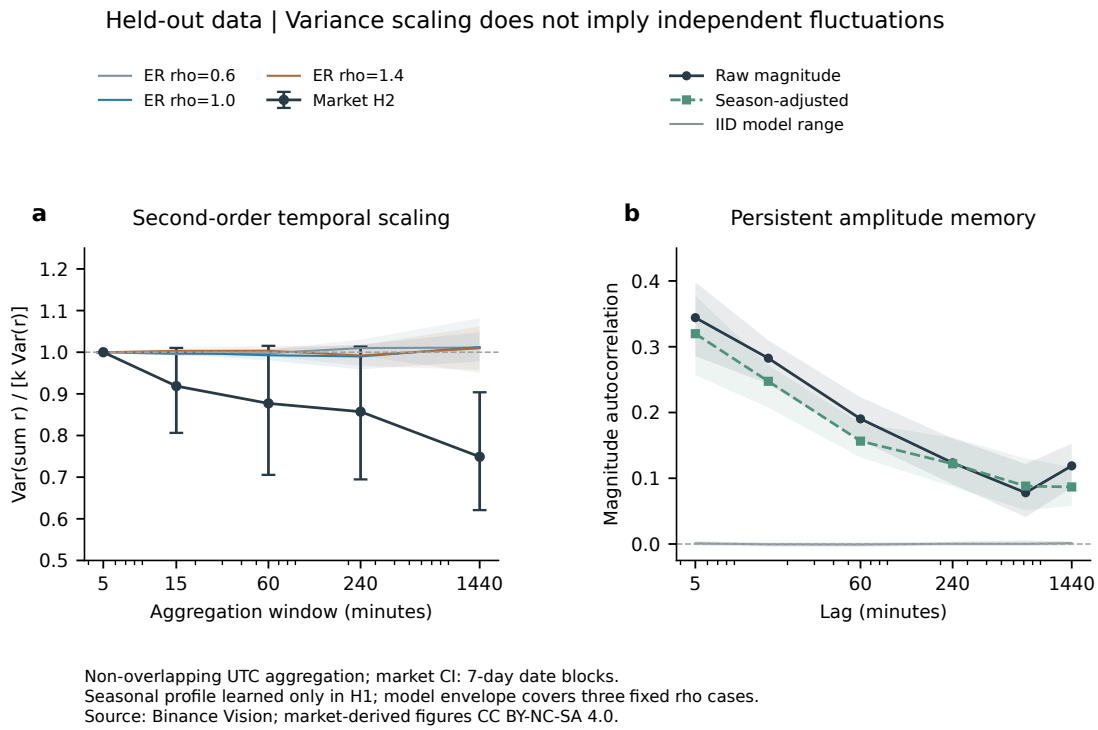


Figure 8. Temporal aggregation and lag dependence in independent model simulations and Bitcoin observations. Market magnitude memory contrasts with the independent-period baseline. Variance ratios and lag correlations describe temporal information, separately from cluster geometry. [R03]

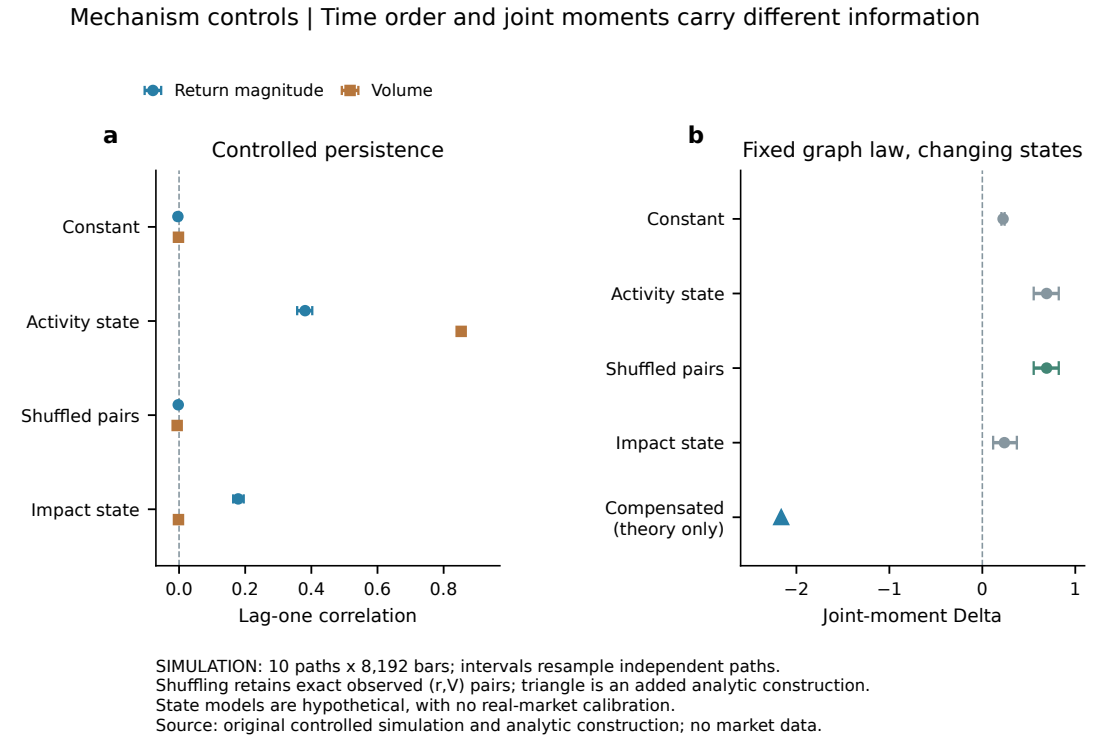


Figure 9. Controlled activity and response states under the same graph law. Persistent states produce different magnitude and volume memory; pair shuffling preserves the empirical same-period joint distribution and removes temporal order. Intervals resample independent paths. Sources: state_mechanism_v2. [V01]

that the positive fixed-interface restriction does not extend to all herding mechanisms. This construction demonstrates how activity–response coupling can reverse the restriction; estimating such a mechanism in market data remains a separate inference task. Full assumptions and exact checks appear in STATE_MECHANISM_v2.md.

The state experiments explain the transition from mechanism analysis to forecasting: an observable may identify a current state, while its predictive value depends on how that state evolves and how the response compensates it. The forecasting benchmark below tests observable temporal information directly.

7 Historical next-day volatility forecasting

7.1 How physics and forecasting are connected

The physics motivates observable activity, joint price–volume statistics, and extreme-concentration measures. Market tests assess the fixed interface and reveal temporal dependence. The forecasting application directly uses these observables and multiscale lags, without true network or cluster labels. The heterogeneous autoregressive (HAR) representation follows the multiscale approach of Corsi[8]. It is an empirical forecasting specification using mechanism-motivated observables and temporal lags, with forecast accuracy evaluated against named statistical references.

After the final five-minute bar of day d closes, the model forecasts the next UTC day:

$$RV_d = \sum_{j=1}^{288} r_{d,j}^2, \quad z_{d+1} = \log RV_{d+1}, \tag{13}$$

$$\widehat{z}_{d+1} = \bar{z}_{\text{fit}} + \widehat{w}^\top \widetilde{x}_d,$$

$$\widehat{w} = \arg\min_w \sum_{d \in \text{fit}} (z_{d+1} - \bar{z}_{\text{fit}} - w^\top \widetilde{x}_d)^2 + \alpha_R \|w\|^2. \tag{14}$$

Standardization uses the fitting period only. The ridge penalty α_R differs from a tail exponent. The displayed $100\sqrt{RV}$ measures intraday volatility intensity, without specifying the direction of the close-to-close return.

Table 3. The full model's ten inputs, all available at the forecast origin.

Group	Definition	Count
HAR	$\log RV_d$; log mean RV over the past 7 days; log mean RV over the past 28 days	3
Activity	Log total daily BTC volume; log ratio of daily volume to its past-7-day mean	2
Coupling	Bar-level Spearman$(\lvert r\rvert, v)$; $\overline{r_c^2 v_c}/(\overline{r_c^2}\,\bar{v})$; absolute summed signed-taker volume divided by daily volume	3
Tail	$\log[288 \sum r_j^4/(\sum r_j^2)^2]$; $\max r_j^2 / \sum r_j^2$	2

r_c, v_c are demeaned within a day. Signed activity derives from venue taker fields, not full order-book OFI or market-wide buy–sell imbalance. Tail inputs measure finite-sample shape and concentration, not power-law exponents. The 1/7/28-day horizons adapt the setup to a 24/7 market.

The original protocol used 337 eligible training targets in 2025 and 90 validation targets in 2026 Q1. After selection, one refit used all 427 targets. Parameters remained fixed across 183 evaluation dates from April 1 to September 30, 2026. Observed test-day data can become later lagged inputs, but do not enter refitting or renewed selection. Figure 10 retains every test date.

7.2 Overall skill and incremental value are separate

The primary loss and relative skill are

$$L_{\log} = D^{-1} \sum_d (\log RV_d - \widehat{z}_d)^2,$$

$$\text{Skill}_B = 100(1 - L_{\text{model}}/L_B). \tag{15}$$

Table 4 and Fig. 11 place simple and strong references together. The full model has positive historical skill over a constant, persistence, and EWMA. HAR alone already improves over persistence by

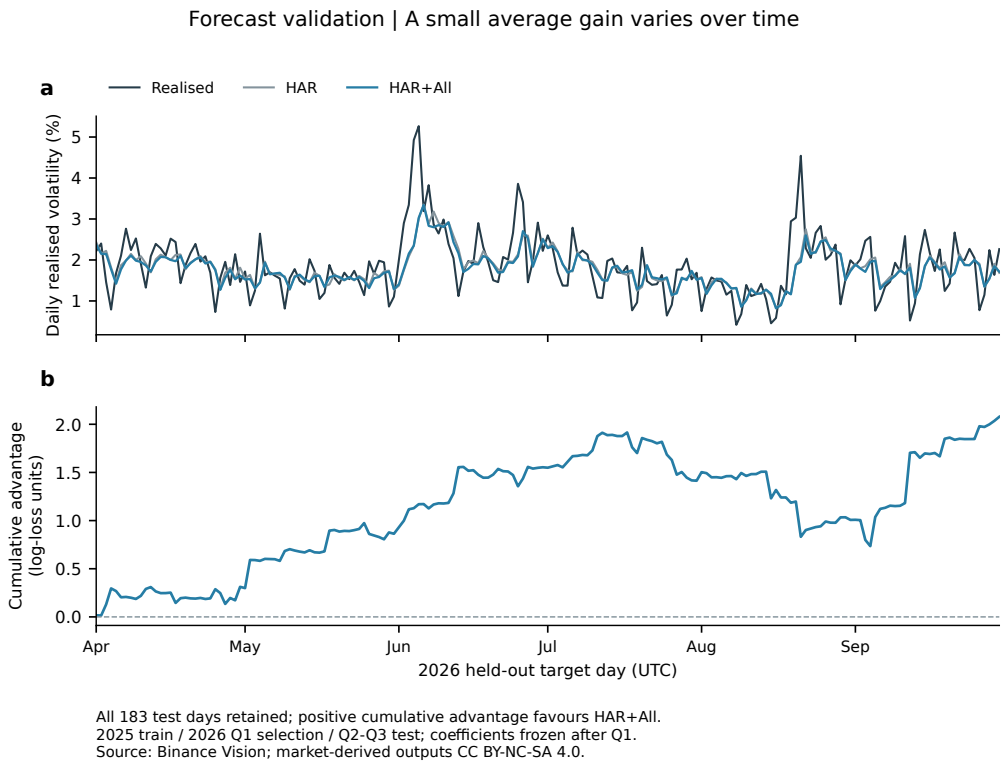


Figure 10. Forecasts and realizations on all 183 historical evaluation dates. Volatility is $100\sqrt{RV}$. The lower panel accumulates forecast-loss advantage over HAR. Binance Vision, CC BY-NC-SA 4.0. [P01]

29.61%. The additional 2.263% point gain over HAR has an interval crossing zero. The original primary comparison has Holm-adjusted $p = 0.3948$, without confirmed incremental advantage. HAR+All adds return shape as well as trading information, so its incremental change is not volume-only.

Table 4. Original next-day log-RV evaluation: 183 BTCUSDT dates.

Reference/model	log-MSE	All loss reduction (%)	Approx. 95% CI (%)
Fit-period mean	0.793999	38.26	[24.65, 48.97]
Persistence	0.712561	31.20	[21.76, 38.62]
EWMA	0.606913	19.23	[13.22, 24.99]
HAR reference	0.501565	2.26	[-0.11, 4.56]
HAR+All	0.490213	—	—

All denotes the original HAR+All. Intervals use 4,999 paired circular seven-day blocks, unadjusted, in a post-analysis overall-skill reassessment. Source: predictive_utility_v1.

This is a reproducible volatility-analysis application with next-day estimates and a complete historical error record. The output is a next-day volatility estimate paired with a date-by-date error record. The additional benefit over HAR is uncertain in this evaluation, and subsequent asset and time tests in Appendix D delimit its generalization. Prospective calibration and decision-specific losses would be needed to assess operational risk use.

8 Portfolio and strategy risk diagnostics

8.1 How much sample risk can common movement conceal?

The US-equity study contains nine sector ETFs, SPY and QQQ references, and 46,530 daily rows through October 23, 2026. Financial association networks have established precedents[9]; our sector graph represents observed co-movement, not trader interactions. The original design uses 2010–2017 for training, 2018–2019 for selection, 2020–2025 for historical evaluation, and 2026 for confirmation.

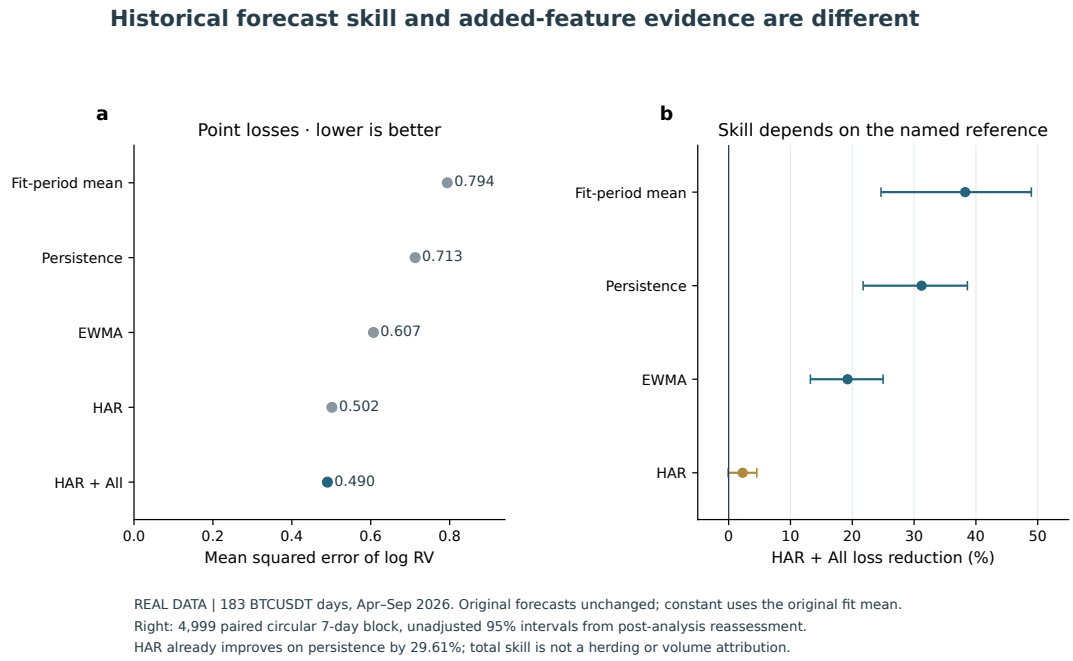


Figure 11. Historical log-MSE and full-model loss reductions against named references. Intervals use 4,999 paired circular seven-day-block resamples. The constant-reference comparison belongs to the post-analysis reassessment; the original HAR comparison is reported in the text. [J02]

For within-window return covariance $\widehat{\Sigma}$ and weights w,

$$\widehat{v} = w^\top \widehat{\Sigma} w = \underbrace{\sum_i w_i^2 \widehat{\sigma}_i^2}_{v_{\text{self}}} + \underbrace{2 \sum_{i<j} w_i w_j \widehat{\sigma}_{ij}}_{v_{\text{collective}}} . \tag{16}$$

Annualized volatility is $100\sqrt{252\widehat{v}}$. For nine equally weighted sectors in the 63-day window ending March 31, 2020, the full estimate is 57.73%. Retaining only diagonal terms gives 20.62%, a ratio of 2.80. Collective terms account for 87.24% of full estimated variance. These are same-window decompositions, not future forecasting gains. Figure 12 retains the longer rolling history.

The existing tool lets users change dates, 63/126-day windows, and holdings. It also evaluates $\Sigma_\eta = (1-\eta)\,\mathrm{diag}(\widehat{\Sigma}) + \eta\widehat{\Sigma}$. The interpolation preserves positive semidefiniteness and gives a conditional covariance scenario for the selected holdings. The interface exposes how estimated common movement changes the same portfolio's risk.

8.2 Strategy variants are not independent risk sources

The six fixed proxies use momentum at 63/126 days, reversal at 5/21 days, and low volatility at 21/63 days. Dependence is assessed primarily through daily net P&L increments. The low-volatility pair has daily correlations 0.921/0.795 in the historical/2026 periods, versus cumulative-curve correlations around 0.982/0.985. Cumulative curves therefore cannot replace daily-risk analysis. Figure 13 separates shared exposures and cross-asset terms.

The default screening rule uses only 2011–2017 daily net increments and turnover. It ranks candidates by turnover and flags positive correlations ≥ 0.80. Six candidates become five representatives, reducing their count by 16.7%. Evaluation-period statistical reconstruction of the flagged strategy gives $R^2 = 83.9\%/64.1\%$. The rule was proposed after evaluation data had been inspected and remains exploratory. Reconstruction R^2 is neither retained alpha nor an executable hedge return.

Table 5 shows higher combined turnover after removing one candidate, together with higher 2026 volatility. The useful workflow flags candidates, then checks combined holdings, costs, and exposures before deciding what to retain. The interface makes all 63 nonempty subsets available for comparing those consequences under a fixed evaluation period.

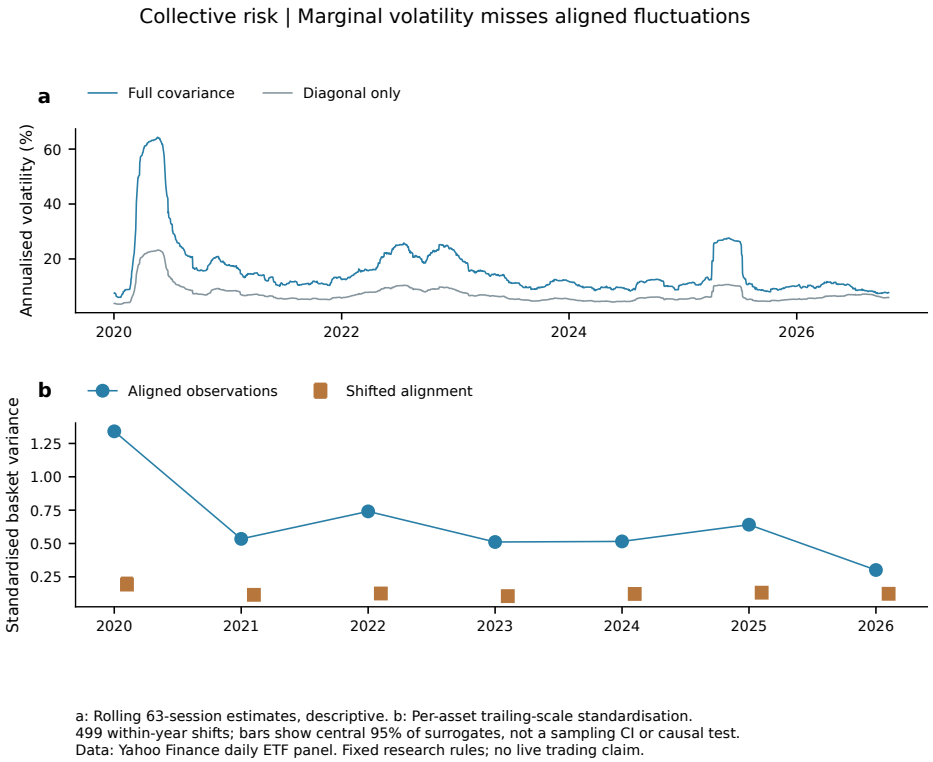


Figure 12. Sector-portfolio risk decomposition and conditional shifts of cross-asset date alignment. Numerical risk uses the full covariance matrix, independently of the plotted edge threshold. Shift ranges describe the conditional alignment comparison. The fixed 2020 case is in risk_lab_v1/risk_snapshots.csv. [U02]

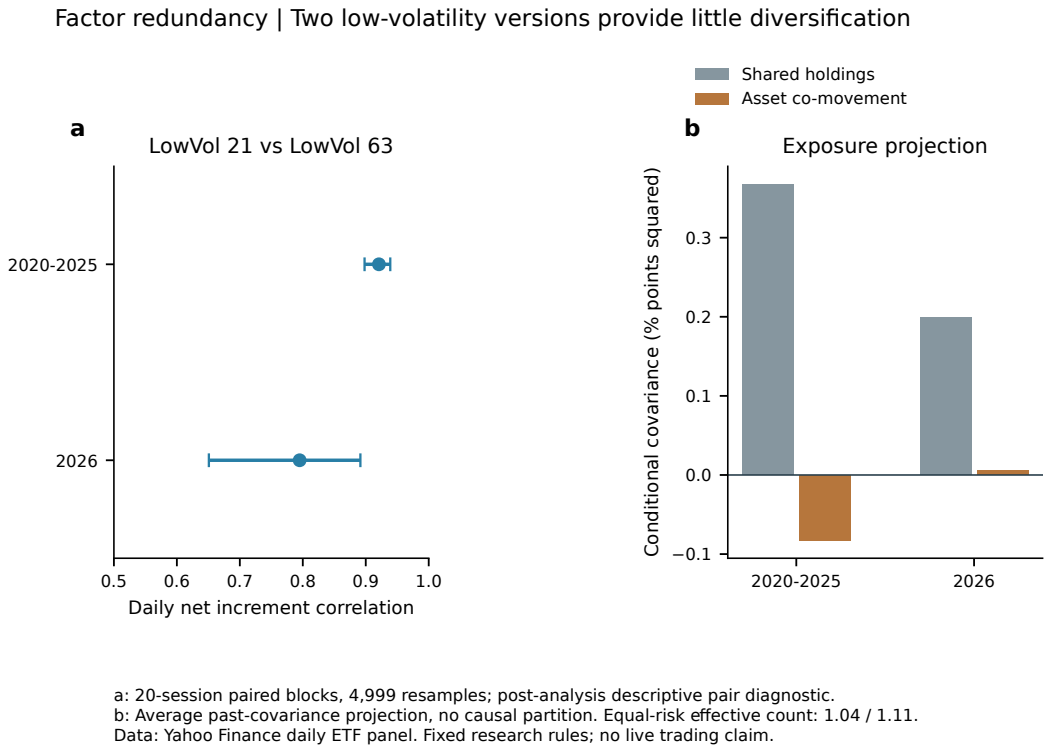


Figure 13. Daily net P&L dependence for the two low-volatility strategy variants, with cumulative-curve comparisons and exposure-based risk decomposition. Intervals use date blocks. For equal-risk strategies, $2/(1+\rho)$ is a covariance-equivalent count. [U05]

Table 5. Default screening from six strategies to five representatives.

Period	Cumulative net P&L (%)	Annual P&L volatility (%)	Daily turnover (%)
2020–2025	-21.68→-20.77	4.35→4.33	7.46→8.67
2026 confirmation	-6.44→-5.57	3.84→3.99	8.03→9.05

Fixed reference capital with additive P&L; 5 bp one-way transaction cost and assumed 100 bp annual borrowing cost. Screening is post-analysis exploration, not a new blind test. Source: risk_lab_v1/subset_metrics.csv.

8.3 How the mechanism enters the tool

With additional assumptions of common loading v, independent noise covariance D, and fixed strategy exposures W,

$$\Sigma_Z = Kvv^\top + D, \quad K = q(\lambda/N)^2 \mathbb{E}S_2,$$
$$\Sigma_{\text{strategy}} = W\Sigma_Z W^\top. \tag{17}$$

This conditional factor projection connects a coordination-generated common mode to the covariance used by the risk tool. Empirical covariance can also reflect common news, holdings, and heteroskedasticity. The projection describes fixed exposures; daily strategy P&L is evaluated separately with its actual time-varying holdings.

The existing single-file tool has four modules: omitted risk, factor screening, allocation comparison, and physical mechanisms. Its mechanism module varies ρ, q, redraws graphs, and simulates actions while displaying cluster moments and model risk. The common-loading scenario is an illustrative model calculation; sector risk uses the observed return covariance. Tool settings, source metrics, and implementation checks accompany the interface.

Source tables and methods: `web/advanced/index.html`.

9 Discussion

9.1 What aggregate observations identify

The analysis separates three questions that are often conflated: recovering parameters within a chosen family, distinguishing microscopic structures, and checking whether the family is compatible with observations. Exact moments support the first task, the characteristic function resolves the chain–star example, and the joint restriction makes the third task operational. This combination places parameter recovery alongside an explicit account of what the observations retain.

The finite ER calculation builds on classical random-graph theory[5], while the calibration problem belongs to the wider study of financial agent-based models[6, 7]. The useful connection here is an analytically specified observation rule whose consequences can be checked at the population, simulation, and market-data levels. Persistent-state experiments sharpen the distinction: changing activity contributes volume variance independently of network geometry, and informative volume can coexist with a constant conditional return-variance target.

9.2 From mechanism tests to risk analysis

The market restriction and the forecast comparison serve different purposes. The former tests a necessary consequence of fixed activity and response; the latter evaluates the historical usefulness of observable lags. Their combination motivates a diagnostic workflow: specify a mechanism, derive its observable restrictions, test those restrictions, and evaluate any proposed predictive features against a strong reference. The uncertainty of the incremental HAR comparison is part of that evaluation.

Portfolio diagnostics extend the same emphasis on collective effects to an observable covariance problem. The risk contribution of co-movement and the overlap between strategy variants are directly computable for selected holdings and dates. Their practical output is an inspectable decomposition and a comparison of costs and exposures. Linking these covariances to a specific microscopic trading mechanism would require additional identifying information.

9.3 Research directions

The next mechanism question is how to distinguish changing activity, changing connectivity, and changing price response in a dynamic setting. The existing state experiments provide controlled alternatives for such a study. Combining them with richer observations could identify which new measurements discriminate between mechanisms. In particular, order-flow imbalance and market depth have an established empirical connection to short-interval price changes[11]; they offer a concrete route for examining the response scale that aggregated bars leave implicit.

Quantum models and quantum machine learning are possible extensions for collective dynamics and inference. A focused first experiment could reuse the known-truth recovery task and compare sample requirements, prediction error, and computational cost with moment inversion and classical learning. Identical classical observation laws remain identical under a common data encoding, so the target must be an identifiable quantity or use additional observations. The role of data and the task-specific nature of quantum learning advantage are central to such comparisons[12]. Whether and when to pursue this direction will depend on problem fit and the value of that comparison.

10 Conclusion

Finite-network moments turn a herding mechanism into explicit statements about observable risk and parameter recovery. Observational equivalence identifies information lost through aggregation, while the joint price–volume restriction supplies a test of the fixed interface in market data. Controlled temporal states explain why current-state information and forecasting information can differ. Together with historical volatility benchmarks and portfolio diagnostics, these results provide a reproducible basis for studying how collective structure enters financial observations and which inferences those observations support.

A Model context: adaptation, bubbles and crashes

The model families in Table 6 address complementary mechanisms. Cont–Bouchaud links random coordination and shared cluster actions to aggregate fluctuations[1]. Our finite implementation specifies both volume observation and unit-order execution. Lux–Marchesi adds switching between opinions and trader types, producing endogenous feedback and temporal structure[2]. The persistent-state experiment in Section 6 isolates temporal activity and response while retaining exogenous state dynamics.

Johansen–Ledoit–Sornette (JLS) relates reinforcing imitation to conditional bubble growth and stochastic crash risk[3]. Its simplified setup is

$$dp/p = \mu(t)\,dt - \kappa\,dj, \quad \mathbb{E}[dj] = h(t)\,dt, \quad \mu(t) = \kappa h(t). \tag{18}$$

The hazard $h(t)$ links growth conditional on survival to compensation for jump risk. The study discusses pre-crash patterns for Wall Street in 1929, 1962 and 1987, and Hong Kong in 1997. These published retrospective cases illustrate the bubble mechanism. The endpoint of conditional bubble growth and the random crash time are distinct model quantities.

Table 6. Mechanisms represented by three financial market models.

Model	Mechanism	Connection to the present analysis
Cont–Bouchaud[1]	Shared actions in random groups	Finite ER model, price–volume moments, inversion and compatibility tests
Lux–Marchesi[2]	Opinion and trader-type switching with profit feedback	Adaptive comparison for the temporal-state experiments
JLS[3]	Reinforcing imitation, hazard and conditional growth	Literature context for bubbles, crashes and criticality

B Complete tail and scaling objects

Cluster number density, node-weighted cluster size, and return distributions are different observables. Figures 14 and 15 retain cluster evidence and separate population size from temporal aggregation. Independent-model second-order aggregation differs from market memory. Similar-looking curves do not establish a universal exponent.

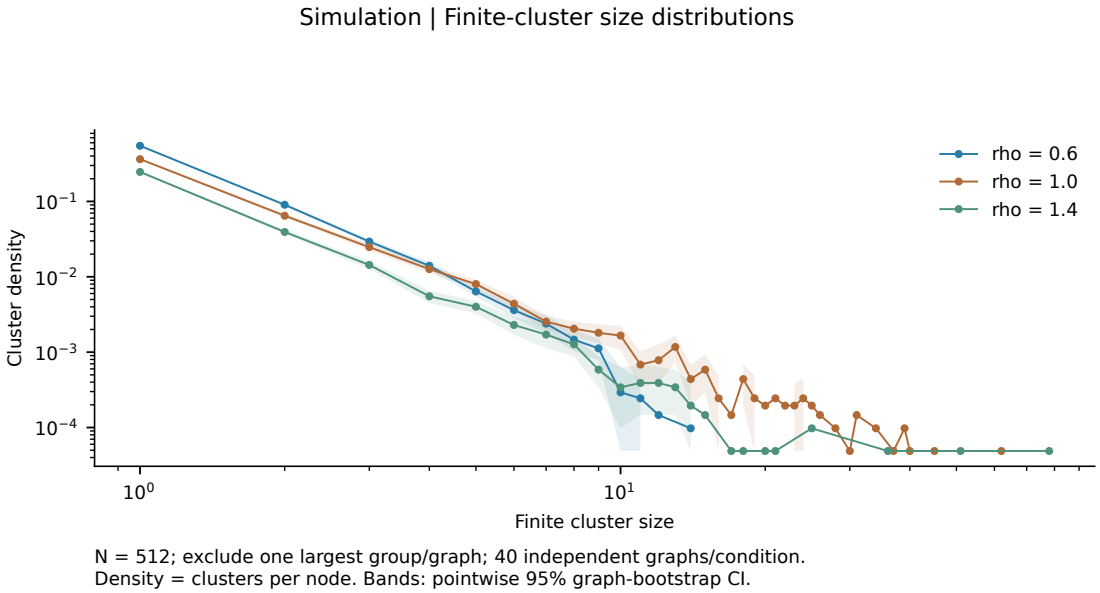


Figure 14. Finite-cluster number density at the specified population size. Pointwise bands resample graphs. Uniform-node sampling would weight each cluster by its size; raw cluster sizes and all population-size source tables are retained. [F04]

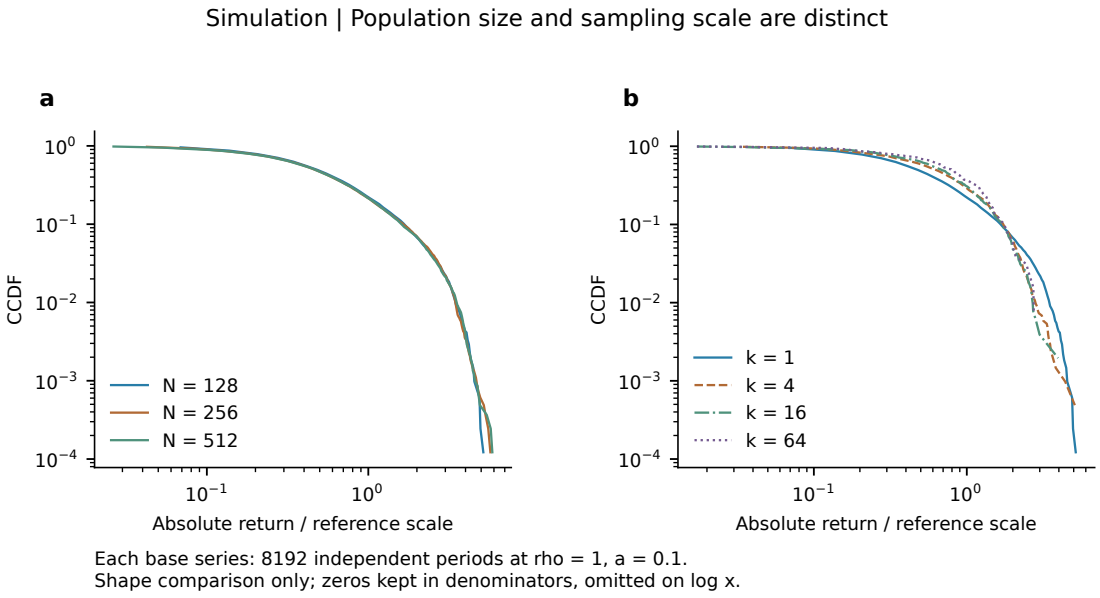


Figure 15. Separate comparisons of population size and temporal aggregation. Return standardization permits shape comparison; temporal blocks are nonoverlapping. The source tables retain all conditions. [F06]

Figure 16 complements joint observations, compatibility, and memory tests rather than selecting only similar-looking tails. All 28 market CSVs and original reproduction records are retained. Source maps distinguish model seeds from observable inputs.

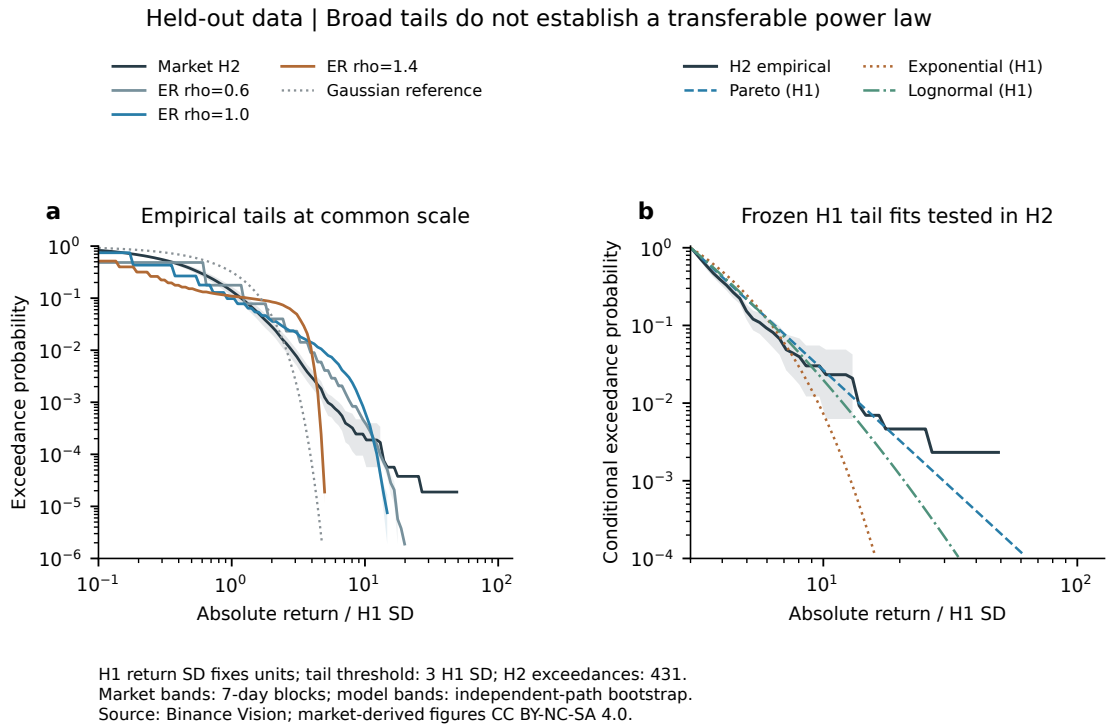


Figure 16. H1-scaled H2 market return CCDFs, conditional tails and continuous alternatives. Zero exceedances remain in the source data and are omitted from logarithmic display. Binance Vision, CC BY-NC-SA 4.0. [R02]

C Exact checks and dependence controls

Theory files contain proofs of Eqs. (2)–(3), monotone identification, and chain/star and mixture equivalence. Rational enumeration and exact certificates check implementations. The original 480 theory checks and ten-study refit records are included. Small-graph enumeration and rational certificates provide independent computational checks of the stated identities. D03 retains the marginal-preserving permutation test: marginal fit cannot replace joint-dependence testing, and that test is not a criticality detector.

D Supplementary forecasts and external evaluations

For a positive realized variance y and positive prediction μ, the quasi-likelihood loss (QLIKE) is

$$\ell_{\mathrm{QLIKE}}(y, \mu) = y/\mu - \log(y/\mu) - 1. \tag{19}$$

QLIKE is a secondary loss in the original log-RV benchmark and the primary loss in the subsequent state-forecast study. The latter's Gamma specifications use a log-linked positive mean and separate penalties on return and trading-information channels. Off denotes the matched return-only model with the trading-information channel fixed to zero. Feature and fitting definitions are in `docs/AB/state_forecast_v2_feature_spec.md`.

Table 7. Supplementary forecast losses, without replacing the primary metric.

Model	QLIKE	Volatility MAE (percentage points)
HAR	0.234214	0.440987
HAR+All	0.233933	0.433735

Volatility is $100\sqrt{RV}$. Exponentiating a log prediction need not yield a conditional arithmetic mean.

Table 7 retains supplementary losses. Monthly point gains over HAR are 2.90%, 5.41%, 4.99%, −1.12%, −1.71%, and 5.15%. A post-analysis comparison against HAR+Tail gives a 1.535% point

gain, again with an interval crossing zero. Only eight high-volatility events occur in the final sample, insufficient for a crash-warning claim.

The subsequent state study adds an external ETH evaluation and 25 new dates for each of BTC and ETH. BTC's new-time primary QLIKE gain is 2.476%, with approximate 95% interval $[-0.407, 4.815]$%. Validation selects Off for the ETH trading channel. Complex Gamma specifications do not consistently beat simple HAR, as retained in Fig. 17. There are 25 distinct dates, not 50 independent observations across two assets.

On the same short BTC period, supplementary log-MSE improves by 3.170% against matched return-only Gamma. Its approximate 95% interval is $[1.236, 4.733]$%, with approximate Holm $p = 0.0276$ across that three-test supplementary family. This positive result does not replace primary QLIKE or external-asset checks, or establish superiority to HAR. Twenty-five dates provide only about 3.6 seven-day blocks, requiring a strong dependence approximation.

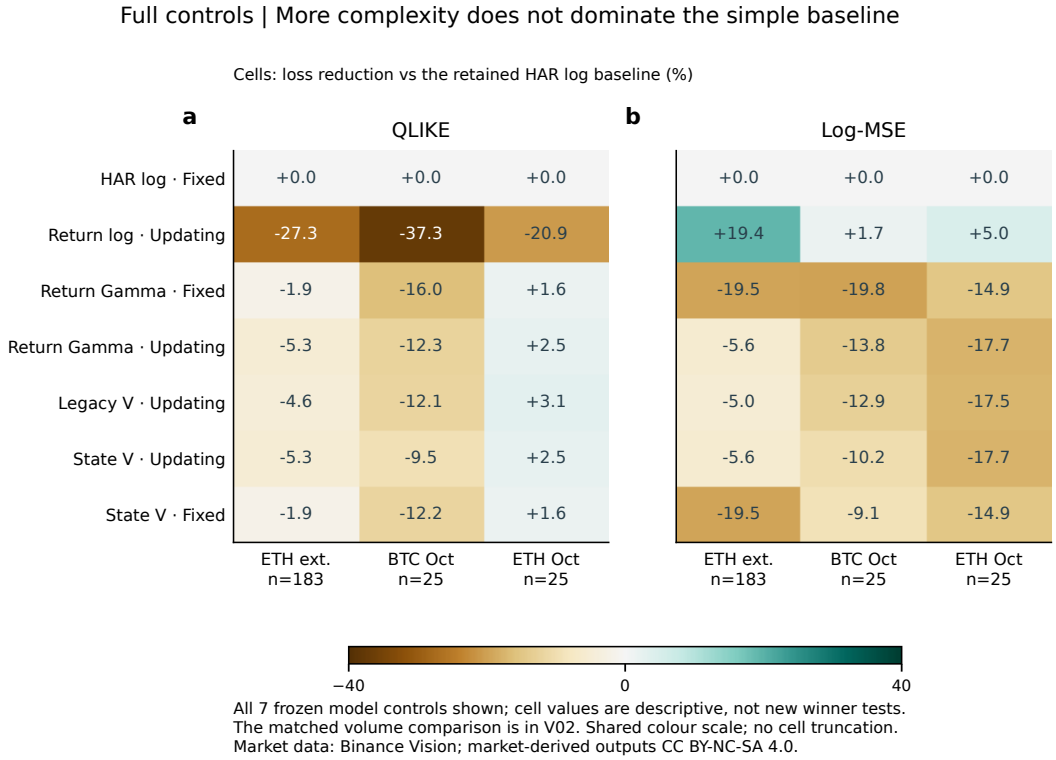


Figure 17. State/price–volume forecasts with simple HAR and matched controls. Previously inspected BTC periods are development information; external-asset and new-time evaluations are reported separately. Off denotes a validation-selected return-only fallback. Binance Vision, CC BY-NC-SA 4.0. [V03]

Covariance-matched tail tests hold information and covariance fixed, comparing nine methods on 38,070 rows from nine external stocks. Historical tail improvement is 0.557%, with Holm $p = 0.0772$; the 2026 point change is -1.797%. There are 32 historical co-crash events and zero in 2026, restricting the latter evaluation to false alarms. Figure 18 and the complete controls do not establish stable market increment. External-asset holdout is not a new future-time blind test.

E Stopped capacity extension and volume trade-offs

Six strategies, nine ETFs, 190 monthly snapshots, four allocations, and three depth levels produce 2,280 conditional states. Volume-only, one-round impact, and multiround-feedback capacity bounds coincide in every state, giving 0% extra reduction in Fig. 19. The maximum initial loss, 1.79158%, is below the fixed 2% trigger. A certificate also verifies nonactivation at smaller scales. The fixed trigger is inactive throughout this design, which makes this branch a certificate of nonactivation under the tested self-impact scenario.

A conventional volume proxy still exposes a useful trade-off. Among 189 Equal-allocation historical replays with three complete future days, mean-volume sizing exceeds the participation proxy 142

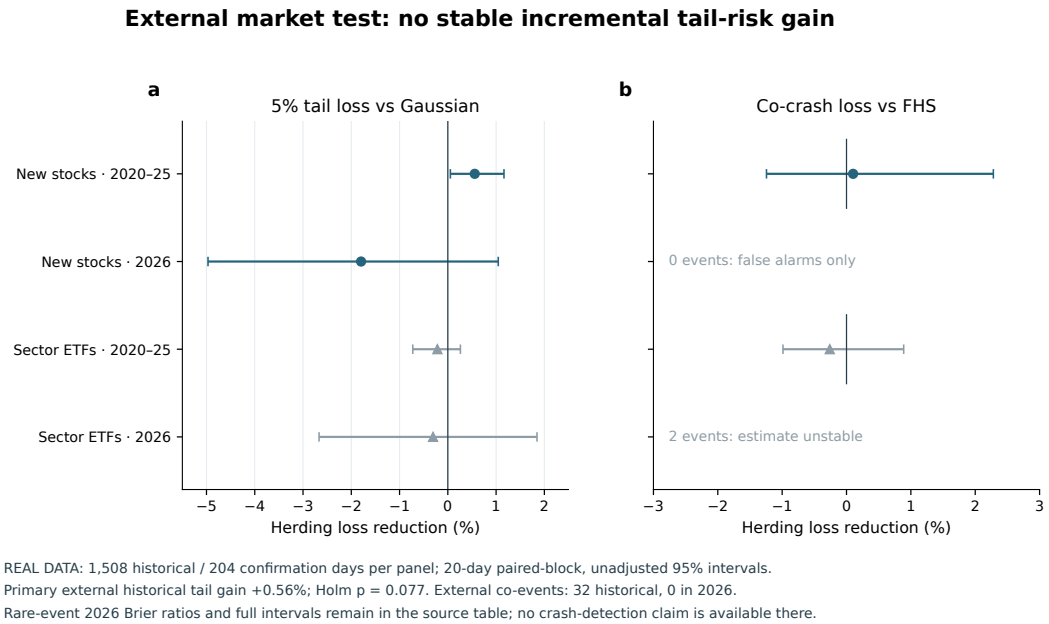


Figure 18. Tail and joint-downside comparisons at matched information and covariance. Statistical references, shrinkage, integration checks and criticality ablations are retained. No co-crash events occur in the 2026 sample, so that period evaluates false alarms. [T03]

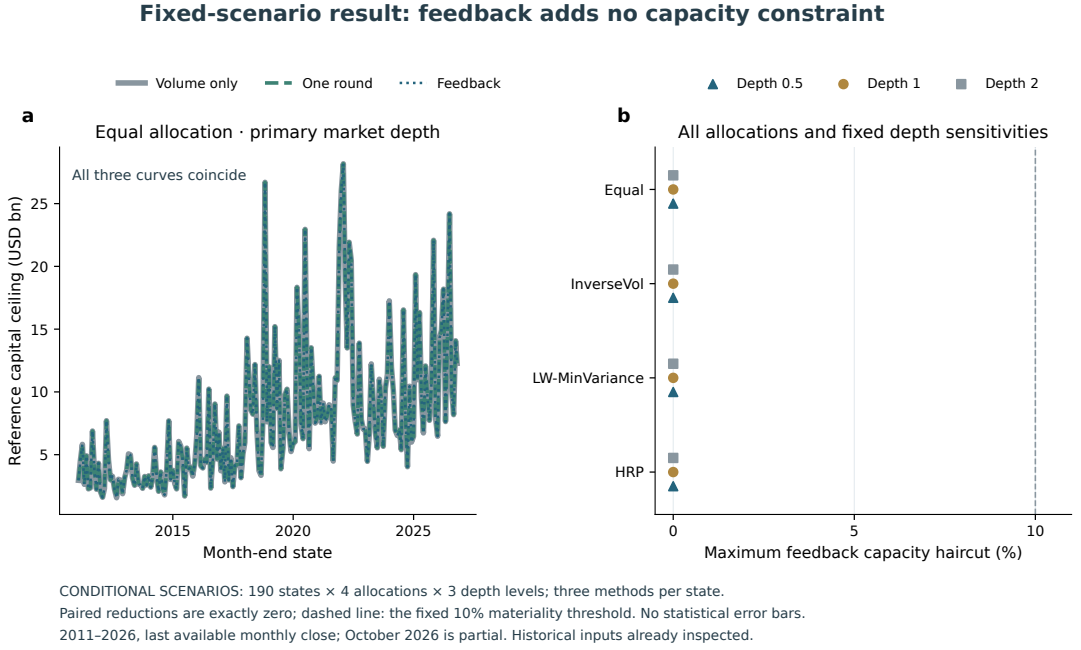


Figure 19. Coincident capacity bounds in the fixed account self-impact scenario. Historical prices and volumes supply conditional inputs; the feedback rule is specified rather than calibrated from execution data. All 2,280 states remain in the source record. [C01]

times, or 75.13%. Using the past-20-day 20th-percentile dollar volume reduces this to 62 cases, or 32.80%, while mean deployed capital falls to 72.90%. This retrospective risk/utilization trade-off is neither a physical-feedback benefit, an actual execution-failure record, nor an optimal-capacity recommendation.

The package includes full code, nine capacity CSVs, cash and monotonicity checks, C01–C03 plotting code, and the original certificate.

F Research record and reproduction

F.1 Finding the evidence

The public repository and research website link the shared implementation in `src/econophysics/`, configurations, numerical tables in `results/tables/`, source manifests and three executable notebooks. Bracketed figure identifiers retain the original F/D/R/P/U/V/T/C/J mapping. Table 8 lists the reproduction entry points.

Table 8. Reproduction profiles from the repository root. Each run writes new outputs.

Profile	Operation	Evidence
quick	Exact theory, model and source-table checks	Standard-library mathematics and numerical identities
theory	Exact identities and certificates	Calculations under the stated model assumptions
notebooks	Execute all three research notebooks	Model moments, inversion, market tests and forecast comparisons
site	Check local resources and the browser engine	Website links and agreement with Python fixtures
forecast	Download documented archives into a fresh directory and rebuild the fixed experiment	All 183 historical test dates and 16 saved CSV tables
Extended laboratory	Rebuild the four-module HTML from retained source and data	Risk, strategy overlap, allocation and coordination diagnostics

F.2 Environment and source records

Forecast reconstruction uses Python 3.14.4 and the versions in `requirements-study.txt`; plotting and notebook dependencies are listed separately. A fresh reconstruction reproduces all 16 saved forecast tables byte for byte. Notebooks retain executed outputs, and the quick and website checks pass on Python 3.11 and 3.14.

Binance Vision-derived results retain CC BY-NC-SA 4.0 attribution. Full supplier price histories are acquired through the source manifests. Code, data and third-party asset terms are documented in `LICENSE.md` and `DATA_LICENSES.md`. Historical experimental digests remain in the source record.

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[10] Binance Vision, Public Market Data; terms and checksum snapshots are included in the package. Market-derived outputs: CC BY-NC-SA 4.0, attribution Binance Vision.

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